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A MACHINE LEARNING PIPELINE DEVELOPED FOR HEART DISEASE PREDICTION

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A MACHINE LEARNING PIPELINE DEVELOPED FOR HEART DISEASE PREDICTION

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ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CAD	Coronary Artery Disease
WHO	World Health Organization
ECG	Electrocardiogram
EDA	Exploratory Data Analysis
FE	Feature Engineering
FN	False Negative
FP	False Positive
GB	Gradient Boosting
GNB	Gaussian Naive Bayes
HPT	Hyperparameter Tuning
IQR	Interquartile Range
IT	Information Technology
LLM	Large Language Model
LR	Logistic Regression
ML	Machine Learning
MLP	Multilayer Perceptron
MSE	Mean Squared Error
NLP	Natural Language Processing
PI	Permutation Importance
RF	Random Forest
SHAP	SHapley Additive exPlanations

ST	ST Segment (in ECG)
TN	True Negative
TP	True Positive
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting
LIME	Local Interpretable Model Agnostic Explanations

ABSTRACT

Nowadays, heart disease is one of the leading causes of death globally and poses major challenges for modern healthcare systems. Specifically, early detection and timely treatment are crucial factors, but people have limitations in accessing this. Therefore, the emergence of ML provides an effective solution to address these challenges by enabling more accurate and reliable disease prediction.

The workflow starts with data preprocessing, followed by a feature engineering (FE) pipeline that includes statistical testing on numerical features, one-hot encoding, statistical testing on categorical (encoded) features, removal of weak features, and feature scaling using StandardScaler. To quantify the impact of FE, we compare two setups: No FE (keeping the original feature set) and FE-enabled (using the refined feature set after weak-feature removal and scaling).

Moreover, six ML models are selected to run simultaneously to compare their disease prediction results: Logistic Regression, Gaussian Naïve Bayes, Random Forest, Gradient Boosting, XGBoost, and Multilayer Perceptron. Then, hyperparameter tuning was applied to improve the performance of each model, and the models were evaluated using accuracy, precision, recall, and F1-score.

Finally, Explainable AI (XAI) methods, including SHAP, LIME, and PI, which are employed to provide transparent and clinically meaningful interpretations. Additionally, insights from PI are used to guide model improvement by generating interaction features, thereby enhancing predictive performance. A Large Language Model (LLM) module is integrated to translate LIME outputs into natural-language explanations, improving interpretability for clinicians and end users.

KeyWords: Heart Disease, Machine Learning, Feature Engineering, XAI, SHAP, LIME, PI, LLM-assisted Explanation.

CHAPTER 1: INTRODUCTION

1.1. Background

Cardiovascular diseases (CVDs) are known as one of the leading causes of death globally, claiming an estimated 17.9 million lives each year [1] . That is an alarming statistic for today's healthcare system. CVDs are types of illnesses related to disorders of the heart and blood vessels, which consist of many disease types, such as coronary heart disease, cerebrovascular disease, rheumatic heart disease, and other diseases. Moreover, it is known that more than four-fifths of CVD deaths result from heart attacks and strokes, with up to a third occurring early in people under 70 years of age [1].

In this study, we focus on one type of CVD, heart disease. Heart disease consists of a wide range of types; some people are born with heart disease and must live with it for the rest of their lives; this is called congenital heart disease. Besides, other types of heart disease develop during a person's lifetime, including coronary artery disease, angina, heart attacks, heart failure, arrhythmia, and cardiomyopathies [2], [3]. Among the types of heart disease, coronary artery disease is assumed to be the most well-known as well as the one causing the most deaths. In this condition, heart problems can occur when plaque builds up in the arteries that supply blood to the heart, causing reduced or blocked blood flow and resulting in insufficient blood supply to the heart. In fact, this type has caused approximately 375,000 deaths [2].

1.2. Problem Statement

Although many heart disease cases can be detected early and treated effectively, current diagnostic methods still face several limitations: high costs, complex procedures, and limited accessibility for large-scale screening. For example, conventional techniques such as angiography, although accurate, are invasive and expensive, making them less suitable for population-wide screening efforts [4]. These limitations hinder early intervention and adversely affect public health outcomes.

Meanwhile, machine learning has shown strong potential in analyzing clinical data to predict disease risk. Recent studies have demonstrated that ML models, including neural networks and ensemble classifiers, can improve early detection of coronary heart disease [5], [6]. However, the performance of these models heavily depends on data quality, preprocessing techniques, and the selection and tuning of algorithms [7]. In practice, many models struggle with consistency and generalizability, particularly when applied to real-world datasets.

To address these challenges, this study aims to build an accurate and easy-to-understand system for predicting heart disease by creating a complete machine learning pipeline. The system includes careful data cleaning, feature engineering, model training with hyperparameter tuning, and Explainable AI methods to show how each feature affects the prediction. In addition, XAI results are used to improve the model further, and an LLM module is added to turn complex explanations into simple natural-language descriptions.

1.3. Scope and Objectives

This study aims to build an effective heart disease prediction model using tabular clinical data and classical machine learning techniques. The scope covers the full machine learning pipeline, including data preprocessing, EDA, feature engineering, model training, hyperparameter tuning, and performance evaluation. First, XAI methods (SHAP, PI) are applied to the model to identify important features. Based on PI results, interaction features is constructed and appended to the feature set; the model is retrained using the same hyperparameters and compared to select the best-performing model for final prediction. After selecting the best model, SHAP and Permutation Importance are used to analyze overall model behavior (global explanations). For individual predictions, LIME is applied to explain each patient case (local explanations). Finally, an LLM module is used to convert LIME outputs into clear, user-friendly explanations.

The dataset consists of commonly available clinical attributes such as age, blood pressure, cholesterol level, heart rate, and lifestyle indicators. Advanced techniques such as deep learning, real-time monitoring, or image-based diagnostics are beyond the scope of this research.

The specific objectives of this study are:

1. Ensure data quality through a data processing pipeline that includes invalid value removal (e.g., cholesterol = 0), column normalization, missing value handling, duplicate value handling, outlier handling, and distribution normalization.
2. Conduct exploratory data analysis (EDA) to visualize data distribution and identify meaningful relationships between features and the target variable.
3. Improve model performance through feature engineering, including numerical feature testing (Welch's t-test), one-hot encoding, categorical feature screening using sample distribution thresholds, weak-feature removal, and feature scaling (StandardScaler); and evaluate its impact by comparing model performance between the original feature set (No FE) and the refined feature set (FE).
4. Develop and compare various machine learning algorithms, including Logistic Regression, Gaussian Naïve Bayes, Random Forest, Gradient Boosting, XGBoost, and Multilayer Perceptron (MLP).
5. Enhance model accuracy and generalization through hyperparameter tuning using techniques such as grid search.
6. Evaluate model effectiveness using classification metrics including Accuracy, Precision, Recall, and F1-score for a comprehensive performance assessment.
7. Identify the most reliable predictive model and assess its potential application in supporting clinical decision-making for early detection of heart disease.

8. Improve the model by creating interaction features from important features baesd on PI's results; the model is retrained and compared to select the best-performing model for final prediction.

9. Use XAI methods (SHAP and PI) to explain the model's overall behavior, and use LIME to explain individual prediction cases. Besides, an LLM is used to turn LIME results into simple, user-friendly explanations for clinicians and non-technical users.

1.4. Research Contributions

The main contributions of this study are summarized as follows:

i. EDA helps identify potential and weak features for the validation and Feature Engineering process. Based on these analyses, a structured Feature Engineering process is applied to evaluate numerical features using statistical testing, transform categorical features using one-hot encoding, remove weak features, and normalize data for model training.

ii. Model Improvement Guided by XAI: Explainable AI techniques are used to analyze the importance and interaction between features after the model is trained. Based on these results, interacting features are constructed and fed back into the training process to improve the model.

1.5. Structure of thesis

This Thesis is comprised of 5 chapters: Introduction, Literature Review, Methodology, Implementation and Results, and Conclusion & Future Work.

Chapter 1 - Introduction

This chapter provides an overview of the research background, defines the problem, research objectives, scope, and presents the overall structure of the thesis.

Chapter 2 - Literature Review

This chapter reviews related work on cardiovascular diseases and machine learning applications in heart disease prediction. It summarizes commonly used algorithms and explainable AI techniques, providing the theoretical foundation and justification for the chosen methods. Besides, previous research on heart disease prediction using machine learning is discussed.

Chapter 3 - Methodology

This chapter describes the proposed research workflow in detail, including data collection, preprocessing, exploratory data analysis, feature engineering, model training, hyperparameter tuning, and evaluation metrics.

Chapter 4 - Implementation and Results

This chapter presents the implementation of the proposed system and reports experimental results. It evaluates model performance using standard classification metrics, compares models with and without feature engineering, assesses generalization on additional datasets, and analyzes model's behavior through XAI techniques. The effectiveness of XAI-guided model improvement and the LLM-based explanation is also discussed.

Chapter 5 - Conclusion and Future Work

This chapter summarizes the main findings of the research, emphasizes the practical implications of the results, and proposes future directions for improving and expanding the study.

CHAPTER 2: LITERATURE REVIEW

2.1.Cardiovascular Diseases

Cardiovascular diseases (CVDs) represent a diverse class of conditions involving the heart and blood vessels, ranging from coronary artery disease and heart failure to cerebrovascular events such as stroke. They are not only the leading cause of death globally but also one of the most persistent burdens on healthcare systems, economies, and quality of life [8]. CVDs often progress insidiously, earning the label of a "silent killer." One of the most prevalent pathological mechanisms is atherosclerosis, characterized by chronic inflammation and lipid accumulation within arterial walls. This process narrows and stiffens the arteries, reducing blood flow and increasing the risk of ischemic events such as myocardial infarction or ischemic stroke [9].

What makes CVDs particularly dangerous is the delay between disease onset and symptom manifestation. Individuals may remain asymptomatic for years, only to experience a sudden, often fatal event. By the time clinical signs such as chest pain, shortness of breath, or neurological deficits emerge, irreversible damage may have already occurred [10]. The societal impact of CVDs extends beyond mortality. They contribute to long-term disability, reduced productivity, and substantial healthcare costs. The World Economic Forum estimates that CVDs alone could cost the global economy over $1 trillion per year by 2030 if current trends continue [11]. Bridging the gap between early detection and intervention through data-driven methods, particularly in low- and middle-income countries where 75% of CVD deaths occur, is not just an innovation - it is a necessity [8].

Cardiovascular diseases (CVDs) often present with acute clinical events such as myocardial infarction or stroke, both of which require immediate medical intervention. The World Health

Organization identifies a core set of symptoms that are commonly associated with each condition and that may serve as early warning signs for timely diagnosis and treatment [8].

Table 2.1 summarizes the typical symptomatology of myocardial infarction and stroke as reported by WHO. These symptoms are critical for both public education and clinical screening protocols, especially in low-resource settings.

Table 2.1: Common Symptoms of Heart Attack and Stroke [8]

Category	**Heart Attack**	**Stroke**
Pain / Discomfort	Central chest pain, pressure, or tightness; may radiate to arm, neck, jaw, or back.	Sudden severe headache without a known cause.
Neurological Signs	Lightheadedness, syncope, or loss of consciousness.	Sudden numbness or weakness, particularly on one side of the body.
Respiratory Symptoms	Shortness of breath.	Trouble with walking, balance, or coordination.
Autonomic Symptoms	Cold sweat, nausea, vomiting.	Sudden confusion, speech difficulty, or trouble understanding.
Visual Disturbances	Less common, but possible if hypotension occurs	Sudden trouble seeing in one or both eyes

2.2. Machine Learning Applications in Heart Disease Prediction

2.2.1. Random Forest

Random Forest (RF) [12] is an ensemble machine learning algorithm that constructs a large number of decision trees during training and aggregates their outputs to make a final prediction.

This technique combines two main strategies - bootstrap aggregation (bagging) and random feature selection - to reduce variance, prevent overfitting, and enhance model accuracy. Figure 2.1 illustrates the idea of a Random Forest.

As a non-parametric and highly versatile classifier, Random Forest has shown exceptional performance in medical data analysis, particularly in cardiovascular disease (CVD) prediction, where data is often high-dimensional, noisy, and nonlinear. The algorithm's ability to capture complex feature interactions and handle imbalanced datasets makes it especially valuable in clinical contexts [13].

In a study by Ahmed and Husien [14], Random Forest was employed in a retrospective cardiology setting to predict disease progression in patients with coronary artery disease. The dataset included variables such as medication history, imaging results, and cardiac biomarkers. Their results demonstrated that the RF-based prognostic model outperformed several other classifiers in identifying patients at high risk of adverse cardiac events, such as heart attacks or strokes. Furthermore, the model proved highly effective in stratifying patients for tailored treatment decisions.

In addition to its predictive accuracy, RF is relatively robust to hyperparameter tuning and can be easily implemented in real-time clinical decision support systems. While the model is less interpretable than simpler algorithms such as logistic regression, it provides feature importance metrics that offer insight into variable relevance, contributing to its partial explainability in practice.

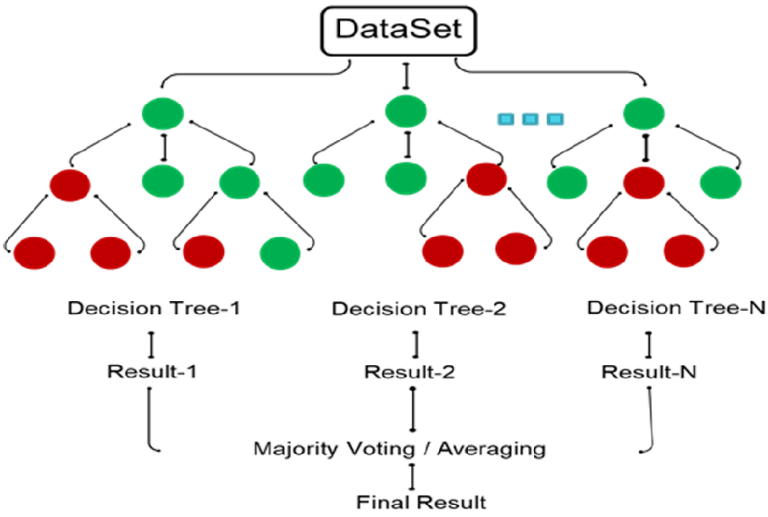

Figure 2.1: Workflow of the Random Forest algorithm [14]

2.2.2. Gradient Boosting

Gradient Boosting (GB) [15] is a versatile machine learning algorithm applied in both regression and classification problems. It functions as an ensemble method where a robust predictive model is developed by combining the outputs of multiple weak learners. The algorithm works iteratively, with each decision tree trained to correct the residuals of the previous model, thus minimizing the loss function in a step-by-step manner. This error-focused learning process enables the model to adaptively improve performance with each iteration. The final prediction is then obtained by aggregating the outputs of all trees in the ensemble.

Hyperparameters such as the number of trees, learning rate, and tree depth can be adjusted to optimize the model's performance. These parameters directly influence the model's complexity, generalization ability, and convergence behavior. As demonstrated in several studies, fine-tuning these settings is essential to prevent overfitting and to ensure efficient learning in real-world datasets.

In practical healthcare applications, GB has been successfully deployed using patient data derived from wearable devices and health monitoring applications. Relevant features such as

sleep patterns, physical activity levels, and heart rate variability were used to train models capable of forecasting the risk of cardiovascular conditions. This proactive prediction enabled timely alerts for patients, encouraging early medical consultation and improving outcomes in preventive cardiology.

2.2.3. XGBoost

XGBoost (eXtreme Gradient Boosting) [16] is an advanced ensemble learning algorithm designed to improve the efficiency and performance of traditional gradient boosting methods. It builds models in an additive manner by sequentially incorporating decision trees, with each new tree focused on correcting the residuals from the previous model iteration. Unlike standard gradient boosting, XGBoost utilizes both the first-order gradient and the second-order derivative (Hessian) of the loss function, enabling more accurate and faster convergence.

A distinctive advantage of XGBoost lies in its regularized objective function, which incorporates both a training loss term and a complexity penalty. This helps reduce overfitting and enhances generalization - an important consideration when working with noisy and high-dimensional clinical data. Additionally, XGBoost supports efficient handling of missing values, sparsity-aware learning, and parallel tree construction, all of which contribute to its computational efficiency.

In the context of cardiovascular disease (CVD) prediction, XGBoost offers several practical benefits. Its ability to model nonlinear feature interactions, robustness to outliers, and interpretability through feature importance rankings make it particularly suitable for structured healthcare datasets. Moreover, its in-built cross-validation and pruning strategies - based on complexity-aware split decisions - allow for more reliable and generalizable models in clinical environments.

2.2.4. Logistic Regression

Logistic Regression [17] is a supervised learning algorithm widely used for binary classification problems, particularly in predicting the likelihood of categorical outcomes, such as the presence or absence of disease. In the context of cardiovascular risk modeling, Logistic Regression can be employed to associate lifestyle and clinical variables with the probability of developing heart disease.

Logistic Regression applies the sigmoid activation function to convert model outputs into probability values. This sigmoid function, characterized by its S-shaped curve, maps any real-valued input into a bounded interval between 0 and 1. As a result, Logistic Regression is particularly useful for estimating the likelihood that an individual may develop cardiovascular disease based on lifestyle-related features.

$$\sigma(x) = \frac{1}{1+e^{-z}} \tag{1}$$

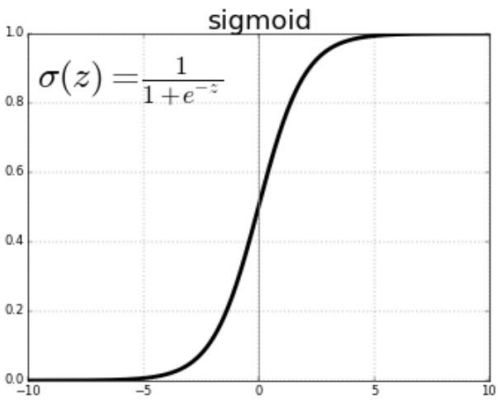

Figure 2.2: Sigmoid Function [17]

In model inference, a threshold (commonly 0.5) is used to convert predicted probabilities into binary labels. If the output $\hat{y} > 0.5$, the instance is classified as positive (e.g., high-risk), otherwise it is negative (e.g., low-risk).

To train the model, the cross-entropy loss function is used:

$$L(\hat{y}, y) = -[\, y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}) \,] \tag{2}$$

This loss penalizes incorrect classifications more heavily as the confidence increases, encouraging the model to make confident and correct predictions. Model parameters are optimized using gradient descent or its variants.

As illustrated in Figure 2.3, input features are combined linearly, passed through a sigmoid function to generate a probability, and evaluated using the cross-entropy loss:

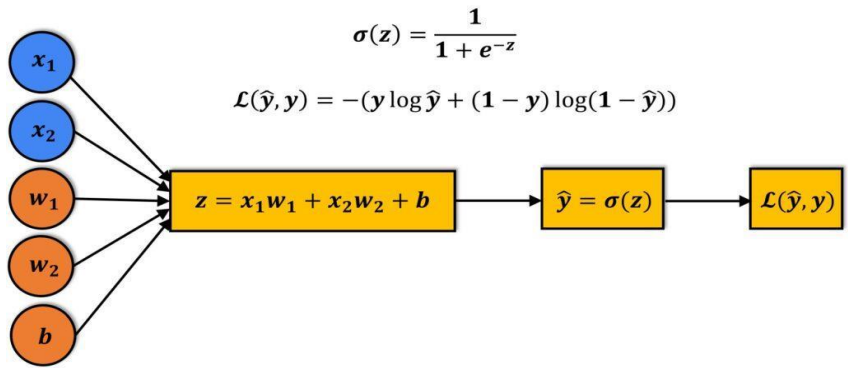

Figure 2.3: Logistic Regression Model Architecture [17]

2.2.5. Gaussian Naive Bayes

Gaussian Naive Bayes (GNB) [18] is a probabilistic machine learning algorithm based on Bayes' Theorem, which assumes that the features used for classification are conditionally independent given the target class. Despite its simplistic assumptions - deemed "naive" in the context of real-world data - GNB has demonstrated robust performance across a range of applications due to its speed, scalability, and effectiveness in high-dimensional spaces.

The "Naive" component of the algorithm refers to its belief that individual features are conditionally independent. For example, in fruit classification, color, shape, and sweetness may all indicate "apple," but the model treats these traits as statistically unrelated. The "Bayes" part is grounded in Bayes' Theorem, which updates prior probabilities using observed evidence, as formulated below:

$$P(A \mid B) = \frac{P\,(B \mid A) \times P(A)}{P(B)} \tag{3}$$

In healthcare applications, particularly heart disease diagnosis, Naive Bayes algorithms have proven effective due to their ability to process structured datasets with multiple risk factors such as blood pressure, cholesterol levels, age, smoking habits, and genetic predisposition. Medical datasets often contain missing values and noise; Naive Bayes' simplicity helps it remain robust under such conditions.

Studies indicate that data mining methods - especially probabilistic classifiers - play a key role in identifying undetected patterns in medical records. By transforming raw medical test data into structured inputs, Naive Bayes can make real-time predictions to assist physicians in early detection and treatment planning. Risk factors such as smoking, physical inactivity, hypertension, high cholesterol, and poor diet are used to estimate the likelihood of conditions such as coronary artery disease, stroke, and peripheral artery disease [18].

2.2.6. Multilayer Perceptron (MLP)

The Multilayer Perceptron (MLP) [19]is a type of feedforward artificial neural network widely used in classification problems, especially in the medical domain where complex, nonlinear relationships exist between clinical features. An MLP consists of an input layer, one or more hidden layers, and an output layer which produces the classification results Figure 2.4. Each neuron in a given layer is connected to every neuron in the subsequent layer, and nonlinear activation functions are applied to introduce learning capacity beyond linear separability.

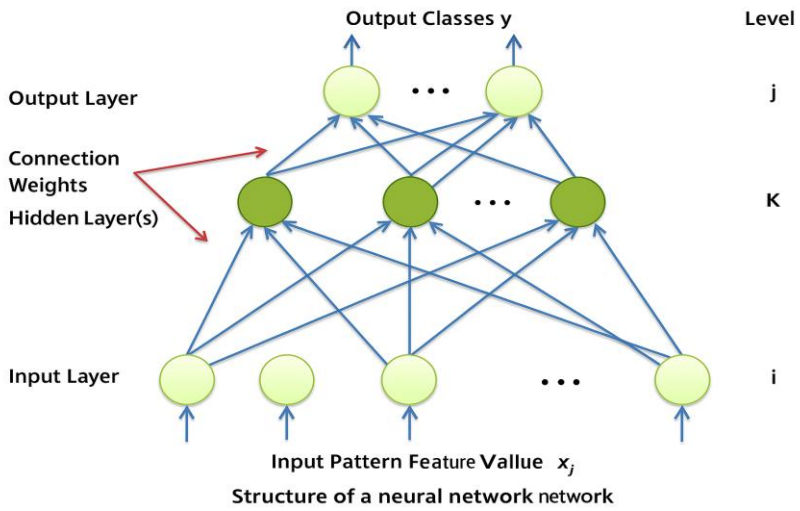

Figure 2.4: Architecture of a MLP Model [19]

In the study conducted by Hasan et al., an MLP was developed for the classification of heart disease using patient datasets sourced from both the UCI repository and local hospitals. The model was trained using the backpropagation algorithm and optimized with the Levenberg-Marquardt technique to minimize the Mean Squared Error (MSE) between predicted and actual labels [19].

The core principle of a neural network lies in its layered computation process. When input data are introduced at the input layer, the interconnected neurons within the hidden layers process the information through successive transformations until a final output is produced at the output layer. This resulting output is expected to indicate the appropriate classification for the given input - ideally, the neuron corresponding to the correct class will exhibit a high activation value, while the others remain comparatively low.

$$y_j = f(\sum_{i=1}^{n} w_{ij}\, x_i + b_j) \tag{4}$$

Each node within a Multilayer Perceptron (MLP) functions as an artificial neuron, as illustrated in Figure 2.6. It calculates the weighted sum of its incoming inputs, incorporates a bias term, and then applies an activation function to generate the final output. This entire mechanism defines the basic computational operation within neural network layers.

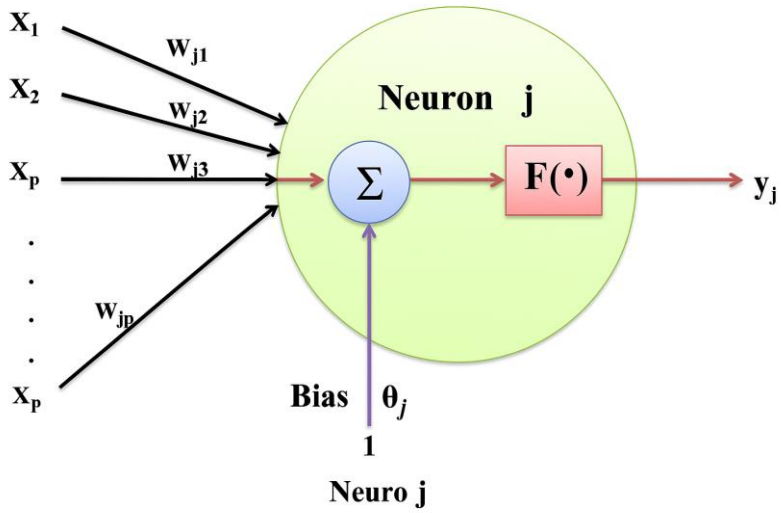

Figure 2.5: One node of MLP [19]

2.3. Explainable Artificial Intelligence (XAI)

Explainable AI (XAI) techniques can be categorized along two main dimensions. The first relates to when interpretability is produced, distinguishing between intrinsic methods, whose transparency is built into the model itself, and post-hoc methods, which generate explanations after the model has been trained. The second dimension concerns the scope of interpretability, which may be global or local.

Global interpretability focuses on understanding the overall behavior and decision logic of the entire model, while local interpretability aims to explain individual predictions — for example, answering the question *"Why did the model produce this specific output?"*. Global explanation techniques help build trust in the model as a whole, whereas local techniques help users gain confidence in a particular decision [20].

In this study, three widely used model-agnostic post-hoc XAI techniques are employed:

- Local Interpretable Model-Agnostic Explanations (LIME) representing the *local post-hoc* category [21].
- SHapley Additive exPlanations (SHAP) representing both *global and local post-hoc* interpretability [22].

- Permutation Importance (PI), which provides global interpretability by estimating the contribution of each feature through measuring the performance drop after random permutation [23]. PI is widely used for model diagnostics and feature selection because it works consistently across any machine learning model.

These methods are chosen because they work with any machine learning model and provide complementary insights into how features influence predictions.

2.3.1. SHapley Additive Explanations

SHAP [24] is a widely recognized method for explaining complex machine learning models, particularly useful in the healthcare domain, such as predicting heart disease. SHAP is rooted in game theory, specifically Shapley values, where each input feature is treated as a "player" contributing to the "outcome" (prediction). SHAP calculates the "fair" contribution of each feature by considering its marginal contribution when it is included or excluded in all possible combinations. This theoretical foundation allows SHAP to offer both local and global interpretability, making it especially valuable in medical contexts. Not only does SHAP help explain why a model predicts heart disease for a particular patient, but it also highlights which features, such as blood pressure, cholesterol levels, and age, are most influential in predicting heart disease overall.

In healthcare, SHAP has been successfully applied to machine learning models based on clinical data for stratifying heart disease risk. For example, clinicians can easily understand how each feature contributes to the prediction, thus increasing trust in decision-support systems. However, while SHAP is a powerful tool, its explanations can be affected by issues like collinearity between features and the complexity of deep learning models, which can lead to misinterpretations if not managed carefully [25], [26]. Despite these challenges, SHAP remains a highly effective tool for ensuring both the accuracy and transparency of heart disease prediction models, thus increasing their reliability in clinical decision-making.

2.3.2. Local Interpretable Model Agnostic Explanations

LIME [27] is another popular method for explaining machine learning models, particularly focusing on local interpretability. LIME generates "perturbed" data by making slight modifications to the features of a given instance (e.g., a patient) and then fits a simpler "glass-box" model, such as linear regression or decision trees, to approximate the decision of the complex model in the local neighborhood. The output of this simpler model serves as the explanation for the original prediction. This process allows LIME to provide transparent and interpretable explanations for individual predictions, which is crucial in healthcare, where clinicians need to understand why a model predicts a high risk of heart disease for a specific patient.

LIME is model-agnostic, meaning it can be applied to any machine learning model, including Logistic Regression, Random Forests, and MLP, making it highly versatile in medical applications. However, LIME's explanations can be less stable due to the random generation of perturbation data, which may lead to different explanations for the same instance on different runs. This introduces a potential issue of "explanation reliability" that must be considered, especially in critical fields such as healthcare [21], [28]). Additionally, the definition of the "neighborhood" around an instance is crucial for LIME’s effectiveness. If the neighborhood is too wide or too narrow, the explanation may not accurately reflect the behavior of the original model.

Despite these limitations, LIME remains a valuable tool for healthcare applications, helping clinicians understand why a model predicts a specific heart disease risk for an individual based on particular features like age, blood pressure, and cholesterol.

2.3.3. Permutation Importance (PI)

Permutation Importance (PI) [23]is a model-agnostic method used to measure how important each feature is for a machine learning model. The idea is straightforward:

- If we randomly shuffle the values of one feature and the model's performance drops a lot, then that feature is important.
- If shuffling the feature does not change the performance much, the feature contributes little to the model.

Because PI does not depend on the internal structure of the model, it can be applied to any type of machine learning algorithm such as linear models, tree-based models, MLP, or ensemble methods. This makes PI a simple and reliable tool for global model interpretation [23].

In healthcare tasks like heart disease prediction, PI helps identify which clinical variables the model relies on the most. For example, shuffling features such as ST slope or oldpeak may cause a strong drop in accuracy, showing that these features play a major role in detecting heart disease. PI can also reveal features that add little or no predictive value, which is useful for feature selection.

However, PI has some limitations. When a feature is shuffled, its correlation with other features is broken, which may cause its importance to be underestimated. In addition, PI requires multiple repeated shuffling steps to obtain stable results, which may increase computation time. Despite these limitations, PI remains a practical and widely used method for understanding global feature importance and complements other XAI techniques such as SHAP and LIME.

CHAPTER 3: METHODOLOGY

3.1. Workflow

In this section, we will present our proposed methodology. This includes key steps from data preprocessing to model evaluation and explainability. In Figure 3.1, the workflow of the proposed system is illustrated, demonstrating the consistent and logical arrangement of the heart disease prediction system throughout a sequence of processing units. The prediction results are able to be explained by popular XAI methods. Additionally, a Large Language Model (LLM) module is integrated to translate these XAI results into clear, human-readable explanations, enhancing interpretability for clinicians and end users.

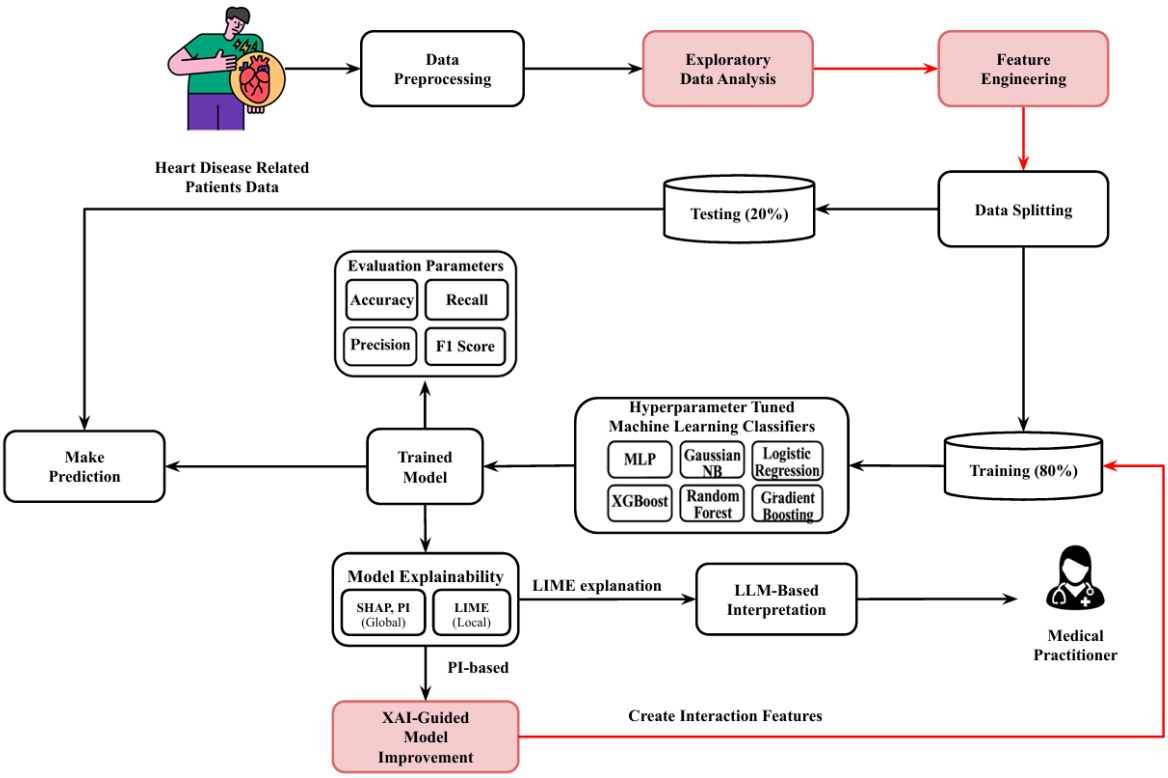

Figure 3.1: Workflow for heart disease prediction using machine learning

3.2. Data Collection

In this study, we selected an available dataset from the Kaggle repository, namely the Heart Disease Dataset, which was published by the user mexwell [29]. The dataset contains 1190 patient records, each including 11 important clinical features such as age, sex, chest pain type,

resting blood pressure, cholesterol level, fasting blood sugar, resting electrocardiogram (ECG), maximum heart rate, exercise-induced angina, oldpeak, the slope of the peak exercise ST segment, and a binary target variable indicating whether a patient has heart disease or not.

3.3. Data Description

This section describes the heart disease dataset used in this study, as shown in Table 3.1.

Table 3.1: The study heart-related dataset features analysis.

No.	Attribute	Code given	Unit Values	Description
1.	age	age	Years	Age of the patient
2.	sex	sex	1 = Male, 0 = Female	Gender of the patient
3.	chest pain type	chest pain type	1, 2, 3, 4	Type of chest pain experienced by the patient
4.	resting blood pressure	resting bp s	mm Hg	Blood pressure measured at rest
5.	serum cholesterol	cholesterol	mg/dl	Serum cholesterol level in the blood
6.	fasting blood sugar	fasting blood sugar	(fasting blood sugar > 120 mg/dl) 1 = true, 0 = false	Fasting blood glucose level
7.	resting electrocardiogram results	resting ecg	0, 1, 2	Resting electrocardiogram (ECG) results
8.	maximum heart rate achieved	max heart rate	71–202	Maximum heart rate achieved during exercise
9.	exercise induced angina	exercise angina	1 = Yes, 0 = No	Chest pain induced by physical activity
10.	oldpeak (= ST depression)	oldpeak	ST depression	ST segment depression induced by exercise
11.	the slope of the peak exercise ST segment	ST slope	1, 2, 3	Slope of the ST segment during peak exercise
12.	class	target	1 = Disease, 0 = Normal	Presence or absence of heart disease

3.4. Data Preprocessing

In this section, we focus on several important preprocessing steps, which includes column normalization, missing value handling, duplicate value handling, outlier handling, and distribution normalization to produce a cleaned dataset, as shown in Figure 3.2.

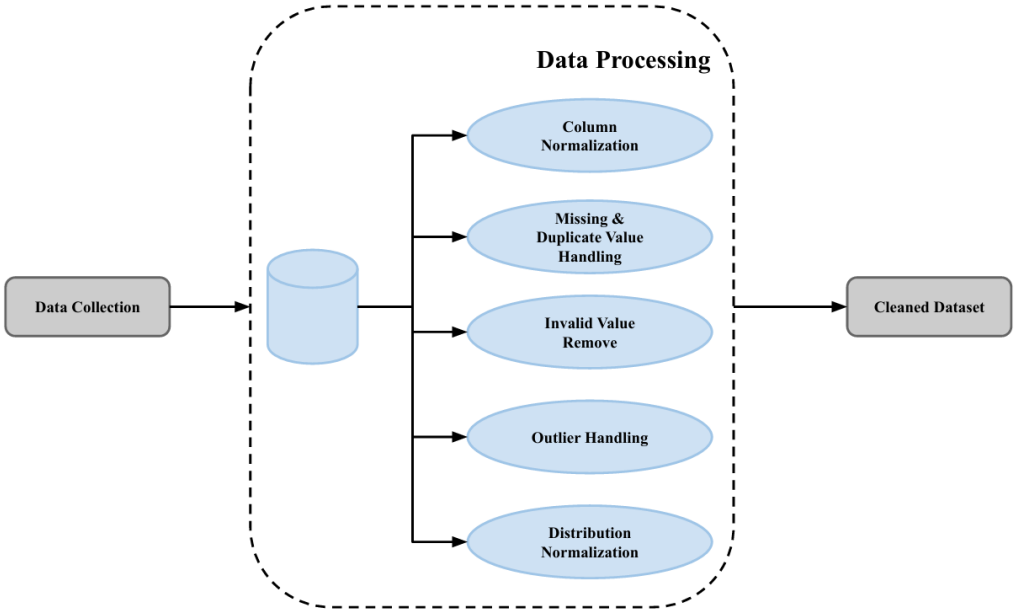

Figure 3.2: Overview of the Data Preprocessing

First, we begin by converting variant column names into a single standardized name to be valid across different data sources in pipeline, as shown in Table 3.2.

Table 3.2: Column Name Mapping and Standardization

Variant Names	Standardized Name
trestbps · resting_bp_s · restingbp · resting_bp · rest_bp · bp	resting bp s
chol · serum_cholesterol · cholestoral	cholesterol
thalach · thalac · max_heart_rate · maxhr · max_hr · heart_rate	max heart rate
cp · chest_pain_type · chestpain	chest pain type
fbs · fasting_blood_sugar · fastingbs	fasting blood sugar
restecg · resting_ecg · rest_ecg	resting ecg
exang · exercise_angina · exer_angina · exerciseangina	exercise angina
slope · st_slope · stslope	ST slope
old_peak · st_depression	oldpeak
num · condition · heart_disease · disease · output · result	target

Next step, we conduct with detecting missing values and duplicate values, which leads us to retain all records in the database because the dataset does not contain any missing or duplicate entries. We also remove invalid values (e.g., cholesterol = 0, oldpeak < 0) to ensure data quality. The next step is to check and handle outliers, since they introduce noise and can negatively affect the performance of ML models. In this study, we applied various methods to handle outliers, including the IQR through histograms and boxplots to assess the distribution and identify potential outliers combined with Winsorization, and logarithmic transformation was applied to reduce skewness where necessary. Specifically, outliers were mainly detected in variables such as resting blood pressure, cholesterol, and oldpeak, which exhibited high variance and asymmetrical distributions. The application of Winsorization effectively limited extreme values without removing valid observations, thereby maintaining data integrity. Afterward, the logarithmic transformation improved the normality of skewed variables, helping to stabilize variance and enhance model learning efficiency. After the preprocessing phase, the data was clean, consistent, and suitable for reliable model training and evaluation, as shown by the more balanced distributions in the numerical features after outlier handling. Figures 3.3 and 3.4 illustrate the data distribution before and after preprocessing, clearly demonstrating the effectiveness of the applied outlier treatment methods.

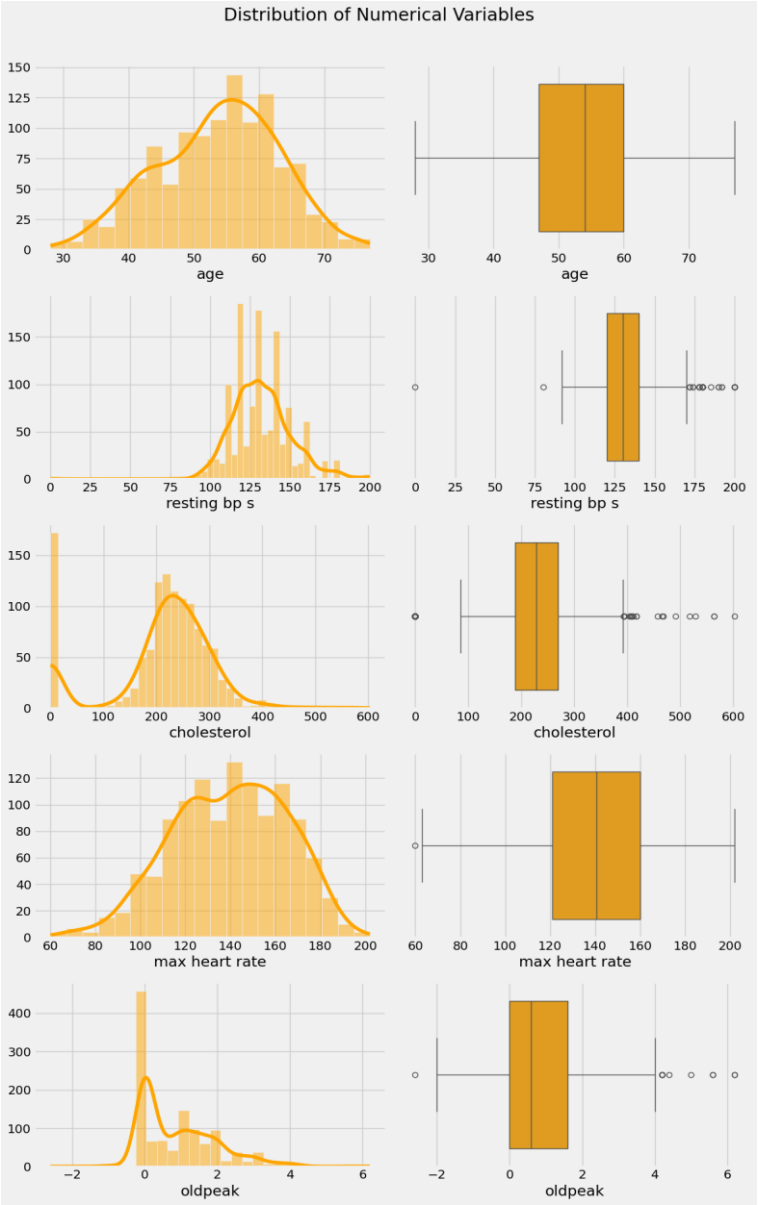

Figure 3.3: Histograms and Boxplots of Numerical Variables

Figure 3.4: Distribution After The Preprocessing Phase

3.5. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was applied to our research dataset to identify valuable statistics. The exploratory data analysis was based on charts and visualizations, from which we extracted meaningful information to support effective technical feature engineering.

The boxplots of numerical features are visualized in Figure 3.5 to analyze potential relationships between features and the target variable.

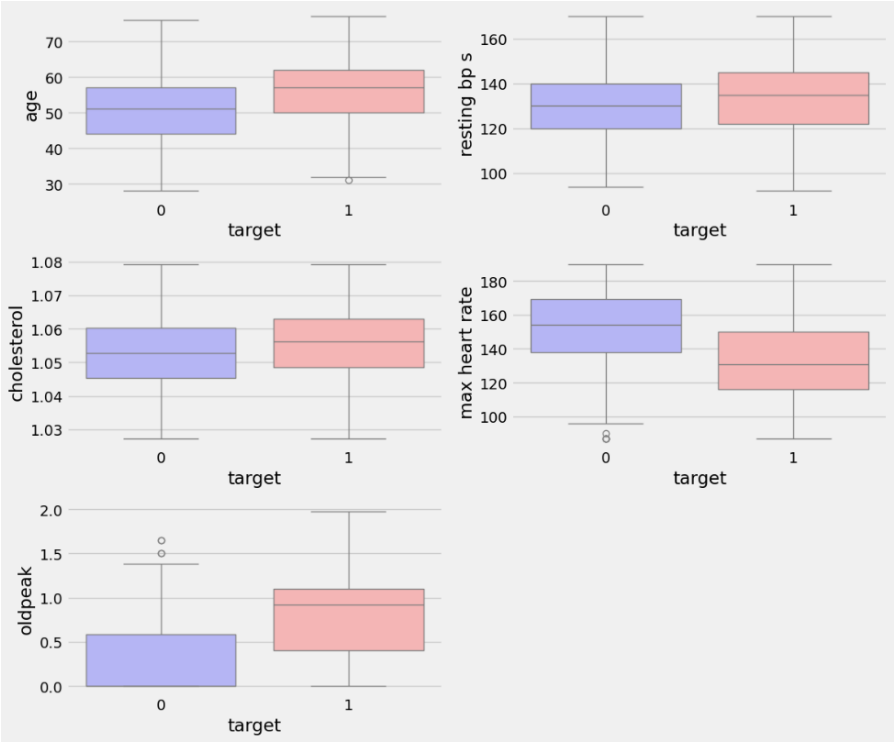

Figure 3.5: Boxplots of Numerical Features by Target Class

The results showed that maximum heart rate and oldpeak exhibited clear differences in distribution between the two groups. Patients with heart disease tended to have lower maximum heart rates and higher oldpeak values, which aligns with well-established clinical expectations. Age was also slightly higher in the heart disease group, as reflected in a higher median value, suggesting that age may serve as a potential risk factor. In contrast, resting blood pressure and cholesterol did not demonstrate clear separation between the two classes based on visual inspection, indicating that these features may have weaker discriminatory power. However, rather than eliminating them during preprocessing, these features were retained for further evaluation using Welch's t-test during the statistical analysis phase. This approach allows us to determine whether the observed differences are statistically significant, thereby informing decisions about whether to exclude features that do not meaningfully contribute to model performance.

Figure 3.6 presents the distribution of samples by age group and gender. The data shows an uneven distribution, with the 44–60 age group having a higher proportion, especially among male, indicating that the most recorded cases are concentrated in this male age group.

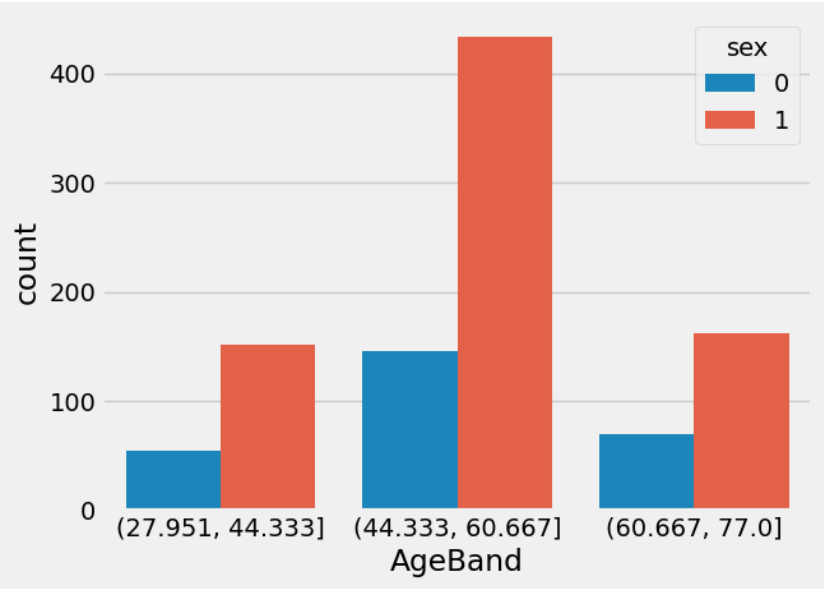

Figure 3.6: Distribution of Patients by Age Band and Sex

Across all categories of ST slope, patients with heart disease (target = 1) generally exhibit higher levels of ST depression (oldpeak) compared to those without the condition. This trend is particularly evident in slope categories 2 and 3, where the median oldpeak values are notably elevated among diseased individuals. Additionally, a positive association can be observed between the ST slope category and the overall distribution of oldpeak, suggesting that steeper ST slopes tend to co-occur with higher degrees of exercise-induced ST depression (Figure 3.7).

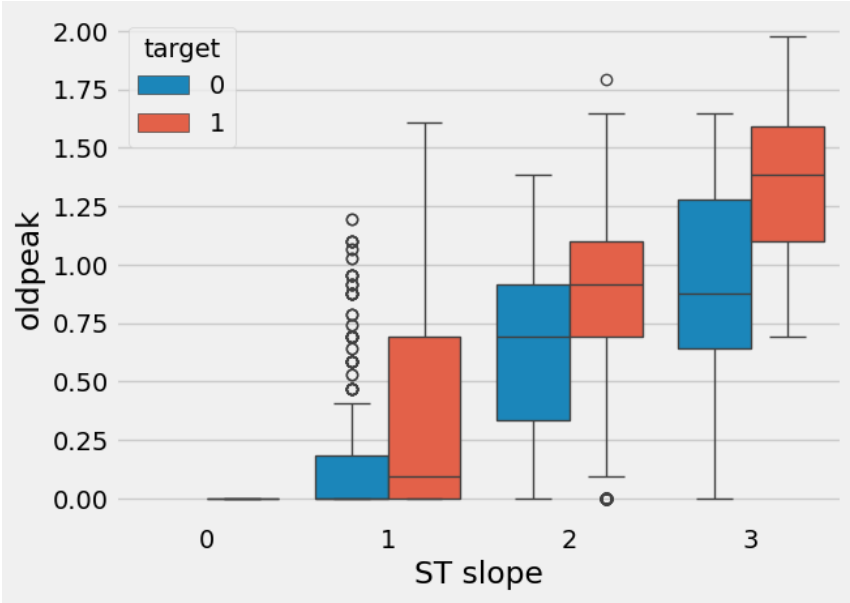

Figure 3.7: Distribution of Oldpeak by ST Slope and Heart Disease Status

These findings imply that oldpeak and ST slope are interrelated features with potential diagnostic value when considered jointly, particularly in the context of identifying ischemic patterns associated with heart disease. Their interaction merits inclusion in the predictive modeling process.

3.6. Feature Engineering

Feature engineering plays a vital role in enhancing the performance of machine learning models by transforming raw data into appropriate and informative input features. In this study, the feature engineering process included several steps, as illustrated in Figure 3.8.

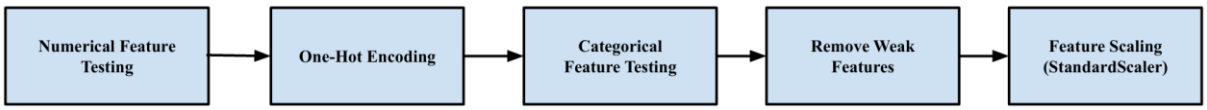

Figure 3.8: Feature Engineering Workflow

Initially, exploratory data analysis (EDA) using boxplots indicated that features such as cholesterol and resting blood pressure did not exhibit distinct separation between the two target classes. Both variables showed similar central tendencies and comparable spread, making it difficult to visually differentiate between the groups. As a result, statistical testing was

conducted to further investigate potential differences in means. Specifically, Welch's two-sample t-test was applied, as it does not assume equal variances - a common violation in biomedical datasets.

The test results in Table 3.2 show the p-values for several numerical features, all of which are less than the 0.05 significance threshold, indicating that each of these features is statistically significant. Specifically:

- Age had a p-value of 0.0000, which is less than 0.05, confirming that this is an important characteristic, and therefore we must retain this feature in the dataset.
- Resting blood pressure (resting bp s) had a p-value of 0.0000, also indicating statistical significance, so it is retained in the dataset.
- Cholesterol showed a p-value of 0.0004, which is still less than 0.05, confirming its statistical significance, so it is kept in the dataset.
- Max heart rate had a p-value of 0.0000, which is below the 0.05 threshold, indicating statistical significance, so we have to keep this figure in the dataset.
- Oldpeak also had a p-value of 0.0000, indicating it is statistically significant and consequently we have to keep it.

Table 3.3: Welch's t-Test Result for each attribute

Feature	**p-value**	**Significance**	**Action**
Age	0.0000	< 0.05 (significant)	KEEP
Resting BP (s)	0.0000	< 0.05 (significant)	KEEP
Cholesterol	0.0004	< 0.05 (significant)	KEEP
Max Heart Rate	0.0000	< 0.05 (significant)	KEEP
Oldpeak	0.0000	< 0.05 (significant)	KEEP

Next, categorical variables with more than two groups such as "chest pain type", "resting ecg", and "ST slope" were processed using one-hot encoding via the pd.get_dummies() function

with dtype = int, converting them into integer-based dummy variables suitable for machine learning models. This encoding process produced new columns such as “chest pain type_1”, “resting ecg_1”, “ST slope_0”, and “ST slope_3”,...

For categorical features, we examined the sample distribution (Table 3.3) and evaluated the significance of each variable. The results are as follows:

Table 3.4: Distribution of Categorical Feature Levels

Feature	Sample Size	Percentage	Action
Chest Pain Type			
chest pain type_1	61 samples	6.0%	KEEP
chest pain type_2	209 samples	20.6%	KEEP
chest pain type_3	248 samples	24.4%	KEEP
chest pain type_4	499 samples	49.1%	KEEP
Resting ECG			
resting ecg_0	577 samples	56.7%	KEEP
resting ecg_1	127 samples	12.5%	KEEP
resting ecg_2	313 samples	30.8%	KEEP
ST Slope			
ST slope_0	1 sample	0.1%	REMOVE
ST slope_1	479 samples	47.1%	KEEP
ST slope_2	476 samples	46.8%	KEEP
ST slope_3	61 samples	6.0%	KEEP

As shown in the analysis:

- Chest Pain Type and Resting ECG have well-distributed samples and are kept in the dataset.

- ST Slope_0 had only 1 sample (0.1%), which is less than 5% of the total dataset. This small sample size raised concerns about its statistical significance, and therefore, it was removed from the dataset.

In addition to statistical testing and sample distribution evaluation, we can observe this more visually through point plots and histogram. Point plots provide an intuitive representation of the mean target value and its associated confidence interval across each category, allowing clearer interpretation of how disease prevalence varies with categorical attributes.

Figure 3.9 illustrates the point plot for the *ST slope* feature. The mean target value for ST slope = 0 appears markedly different from the other categories; however, the absence of a confidence interval (error bar) is immediately noticeable. This occurs because the category contains only one observation, resulting in a confidence interval of zero width. Consequently, this category does not offer any statistically reliable estimate of the true underlying mean. The plot therefore visually reinforces the earlier conclusion that ST slope = 0 lacks sufficient sample representation and should be removed from the dataset.

In contrast, categories ST slope = 1, 2, and 3 display clearly defined mean target values along with non-zero confidence intervals. These intervals indicate varying levels of uncertainty across categories:

- ST slope = 1 and 2 exhibit narrow confidence intervals, suggesting stable and consistent target distributions, thus supporting their retention as meaningful predictors.
- ST slope = 3, although characterized by a wider confidence interval, still contains a reasonable number of samples (61 observations), making it statistically interpretable. The wider interval reflects higher variability but does not warrant removal.

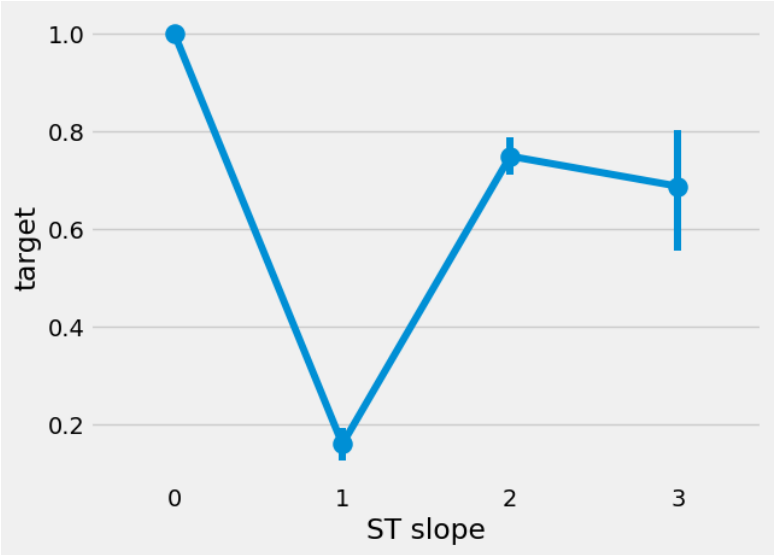

Figure 3.9: Point Plot of ST slope vs. Heart Disease Target with Dispersion

This method helped to achieve high prediction accuracy for heart disease prediction in this study.

3.7. Data Splitting

To evaluate the performance of the model, the dataset was divided into two parts: 80% for training and 20% for testing. The training set was used to learn patterns in the data, while the test set was retained to evaluate prediction accuracy on unseen data. This also helped the models avoid overfitting and obtain more reliable results.

3.8. Evaluation Methods

In this study, we evaluated the performance of each model using the confusion matrix with regard to Accuracy, Precision, Recall, and F1-Score [30].

Confusion Matrix: a square matrix of size N × N, where N denotes the number of output classes, providing information on the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN); this forms the basis for calculating important evaluation metrics.

Accuracy: evaluates overall model performance by measuring the number of correct predictions across the entire dataset.

$$Accuracy = \frac{(\mathrm{TP} + \mathrm{TN})}{(TP + FP + TN + FN)} \tag{5}$$

Precision and Recall: Precision reflects the proportion of correct positive predictions out of the total number of positive predictions, while Recall represents the ability of the model to detect true positive cases.

$$Precision = \frac{\mathrm{TP}}{(TP + FP)} \tag{6}$$

$$Recall = \frac{\mathrm{TP}}{(TP + FN)} \tag{7}$$

F1-Score: is used to balance precision and recall, which helps to evaluate the model more fairly.

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{7}$$

CHAPTER 4: IMPLEMENTATION AND RESULTS

4.1. Model Implementation

Prior to model training, the dataset was partitioned into feature variables X and target labels y, with an 80:20 train-test split using grouped sampling by strata to preserve class distribution. A fixed random seed (42) was applied to ensure reproducibility. To normalize feature scales, the training data were standardized with StandardScaler, and the learned parameters were subsequently applied to the test set to avoid data leakage.

A general-purpose evaluation function, evaluate_model(), was implemented to assess classifier performance using Accuracy, Precision, Recall, F1-score, and Confusion Matrix. Predictions were generated on both training and test sets, enabling consistent comparison across models.

Six machine learning classifiers were selected to represent diverse modeling perspectives. Logistic Regression served as a strong linear baseline; Random Forest and Gradient Boosting were included for their ability to capture non-linear relationships; Gaussian Naive Bayes offered a simple probabilistic alternative; the Multilayer Perceptron (MLP), a feed-forward neural network, provided a non-linear learning approach; and XGBoost was chosen as a powerful boosted ensemble method. All models followed the same preprocessing, training, and evaluation pipeline to ensure methodological consistency, replicability, and a fair comparative analysis.

4.2. Hyperparameter Tuning

To enhance predictive performance, the models were tuned using GridSearchCV with stratified 5-fold cross-validation, where the F1 macro-score served as the optimization metric to provide a balanced evaluation of precision and recall in the binary classification task. Each classifier was assigned a tailored parameter grid (e.g., C for Logistic Regression, and the number of estimators and maximum tree depth for ensemble-based methods). The best configuration was

selected based on the highest cross-validated score, after which the optimized models were re-evaluated on the test set to ensure fair comparison.

4.3. Experimental Results

This section presents the experimental results obtained from the machine learning models after hyperparameter tuning. To ensure a fair comparison, all models were first optimized using the same hyperparameter tuning strategy and then evaluated under two conditions: without Feature Engineering (No FE) and with Feature Engineering (FE).

4.3.1. Performance of Tuned Models (No FE & FE)

4.3.1.1. Not Applying Feature Engineering

As shown in Table 4.1, all tuned models achieve relatively high performance on the original dataset without feature engineering. The best-performing model in this setting is the MLP, with an accuracy and F1-score of 0.9363, followed closely by Gradient Boosting (accuracy = 0.9314) and XGBoost (accuracy = 0.9265). Gaussian Naïve Bayes and Logistic Regression obtain lower but still acceptable scores, reflecting their simpler model capacity compared to ensemble-based methods and MLP.

Table 4.1: Performance Comparison of Tuned Models Without Feature Engineering

Model	Accuracy	Precision	Recall	F1-score
MLP	0.9363	0.9363	0.9362	0.9363
GaussianNB	0.8235	0.8245	0.8238	0.8235
LogisticRegression	0.8382	0.8389	0.8385	0.8382
XGBoost	0.9265	0.9273	0.9267	0.9265
RandomForest	0.9118	0.9123	0.9119	0.9118
GradientBoosting	0.9314	0.9314	0.9314	0.9314

4.3.1.2. Applying Feature Engineering

After applying feature engineering, the overall performance pattern changes (Table 4.2). Gradient Boosting becomes the best-performing model, achieving the highest accuracy and F1-score of 0.9461. XGBoost also benefits from feature engineering, with its accuracy increasing from 0.9265 to 0.9363. Random Forest and Gaussian Naïve Bayes show slight improvements, whereas MLP and Logistic Regression experience a small decrease in performance. Overall, feature engineering improves the performance of 4 out of 6 models, particularly tree-based ensemble methods. These results indicate that the proposed statistical feature engineering strategy is particularly effective for tree-based ensemble models, leading to more accurate and stable heart disease prediction.

Table 4.2: Performance Comparison of Tuned Models With Feature Engineering

Model	Accuracy	Precision	Recall	F1-score
MLP	0.9265	0.9265	0.9265	0.9265
GaussianNB	0.8284	0.8284	0.8285	0.8284
LogisticRegression	0.8333	0.8335	0.8332	0.8333
XGBoost	0.9363	0.9365	0.9364	0.9363
RandomForest	0.9216	0.9217	0.9217	0.9216
GradientBoosting	0.9461	0.9461	0.9461	0.9461

4.3.1.3. Performance on Additional Datasets

In this section, we evaluate the performance of various machine learning models on two datasets: Cardiovascular Disease Dataset and Heart Disease Cleveland. The models are assessed using Accuracy, Precision, Recall, and F1-score under two conditions: without Feature Engineering (Not FE) and with Feature Engineering (FE). Moreover, all models have been tuned for optimal performance in both conditions.

Performance on Cardiovascular Disease Dataset

This dataset is collected on Kaggle platform [31] with 1000 samples and 14 attributes.The target variable indicates whether a patient has cardiovascular disease. Before model training, the dataset is cleaned and standardized to ensure consistent input across different data sources.

- **Feature Engineering:**

Based on Welch's t-test is shown in 4.3, age did not show a statistically significant difference between the two classes ($p = 0.7217$) and was therefore excluded.In contrast, cholesterol, max heart rate, oldpeak, and resting blood pressure were all statistically significant ($p < 0.05$) and were retained for model training.

Table 4.3: Numerical Feature Significance on the Cardiovascular Disease Dataset

Feature	**p-value**	**Significant**	**Action**
Age	**0.7217**	**≥ 0.05 (not significant)**	**REMOVE**
Cholesterol	0.0000	< 0.05 (significant)	KEEP
Max Heart Rate	0.0000	< 0.05 (significant)	KEEP
Oldpeak	0.0005	< 0.05 (significant)	KEEP
Resting BP (s)	0.0000	< 0.05 (significant)	KEEP

The table 4.4 shows the results of categorical feature distribution testing. Almost all features are kept because they meet the required conditions, while chest_pain_type_3 is removed due to its low sample proportion.

Table 4.4: Categorical Feature Distribution on the Cardiovascular Disease Dataset

Feature	Sample Size	Percentage	Action
Chest Pain Type			
chest pain type_1	206 samples	21.8%	KEEP
chest pain type_2	293 samples	30.9%	KEEP
chest pain type_3	**43 samples**	**4.5%**	**REMOVE**
chest pain type_4	405 samples	42.8%	KEEP
Resting ECG			
resting ecg_0	439 samples	46.4%	KEEP
resting ecg_1	324 samples	34.2%	KEEP
resting ecg_2	184 samples	19.4%	KEEP
ST Slope			
ST slope_0	180 samples	19.0%	KEEP
ST slope_1	291 samples	30.7%	KEEP
ST slope_2	294 samples	31.0%	KEEP
ST slope_3	182 samples	19.2%	KEEP
CA			
ca_0	268 samples	28.3%	KEEP
ca_1	325 samples	34.3%	KEEP
ca_2	248 samples	26.2%	KEEP
ca_3	106 samples	11.2%	KEEP

Table 4.5 shows the performance of different models with and without feature engineering. The results indicate that feature engineering improves the performance of four out of six models, including MLP, Gaussian Naive Bayes, Logistic Regression, and Gradient Boosting.

In contrast, Random Forest shows no change, while XGBoost slightly decreases in performance. Overall, feature engineering has a positive impact on most models.

Table 4.5: Model Performance on the Cardiovascular Disease Dataset

Model	**Not Feature Engineering**				**Feature Engineering**			
	Accuracy	**Precision**	**Recall**	**F1-score**	**Accuracy**	**Precision**	**Recall**	**F1-score**
MLP	0.9632	0.9617	0.9631	0.9624	0.9789	0.9773	0.9801	0.9785
GaussianNB	0.9263	0.9260	0.9231	0.9244	0.9474	0.9477	0.9446	0.9460
LogisticRegression	0.9579	0.9552	0.9601	0.9572	0.9632	0.9617	0.9631	0.9624
XGBoost	0.9895	0.9880	0.9908	0.9893	0.9842	0.9832	0.9847	0.9839
RandomForest	0.9789	0.9773	0.9801	0.9785	0.9789	0.9785	0.9785	0.9785
GradientBoosting	0.9842	0.9821	0.9862	0.9839	0.9895	0.9892	0.9892	0.9892

Performance on Heart Disease Cleveland

The dataset is collected from a publicly available source on Kaggle [32]. This dataset contains 297 examples and 14 attributes. The target value is binary, which demonstrates whether heart disease is present or not.

- **Feature Engineering:**

As shown in Table 4.6, the Welch's t-test indicates that cholesterol is not statistically significant ($p = 0.0917$), so this figure is removed from the feature set. The remaining numerical features, including age, max heart rate, oldpeak, and resting blood pressure, show significant differences between the two classes ($p < 0.05$) , this is reason to keep them for model training.

Table 4.6: Numerical Feature Significance on the Heart Disease Cleveland Dataset

Feature	p-value	Significant	Action
Age	0.0001	< 0.05 (significant)	KEEP
Cholesterol	**0.0917**	**≥ 0.05 (not significant)**	**REMOVE**
Max Heart Rate	0.0000	< 0.05 (significant)	KEEP
Oldpeak	0.0000	< 0.05 (significant)	KEEP
Resting BP (s)	0.0113	< 0.05 (significant)	KEEP

The table 4.7 presents the results of categorical feature distribution testing. Most feature levels satisfy the required conditions and are they are retained in the dataset, while resting ecg_1 is removed due to its low sample proportion.

Table 4.7: Categorical Feature Distribution on the Heart Disease Cleveland Dataset

Feature	Sample Size	Percentage	Action
Chest Pain Type			
chest pain type_0	23 samples	7.7%	KEEP
chest pain type_1	49 samples	16.5%	KEEP
chest pain type_2	83 samples	27.9%	KEEP
chest pain type_3	142 samples	47.8%	KEEP
Resting ECG			
resting ecg_0	147 samples	49.5%	KEEP

resting ecg_1	**4 samples**	**1.3%**	**REMOVE**
resting ecg_2	146 samples	49.2%	KEEP
ST Slope			
ST slope_0	139 samples	46.8%	KEEP
ST slope_1	137 samples	46.1%	KEEP
ST slope_2	21 samples	7.1%	KEEP
CA			
ca_0	174 samples	58.6%	KEEP
ca_1	65 samples	21.9%	KEEP
ca_2	38 samples	12.8%	KEEP
ca_3	20 samples	6.7%	KEEP
Thal			
thal_0	164 samples	55.2%	KEEP
thal_1	18 samples	6.1%	KEEP
thal_2	115 samples	38.7%	KEEP

As shown in Table 4.8, FE improves the performance of 5 out of 6 models, which includes MLP, Gaussian Naive Bayes, Logistic Regression, Random Forest, and Gradient Boosting, while XGBoost experiences a slight decrease. Overall, the results indicate that feature engineering is beneficial for most models.

Table 4.8: Model Performance on Heart Disease Cleveland

Model	Not Feature Engineering				Feature Engineering			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
MLP	0.7333	0.7366	0.7366	0.7333	0.7667	0.7760	0.7723	0.7664
Gaussian NB	0.7167	0.7286	0.7232	0.7160	0.7833	0.7825	0.7835	0.7828
Logistic Regression	0.7333	0.7321	0.7321	0.7321	0.7833	0.7848	0.7857	0.7833
XGBoost	0.7500	0.7492	0.7500	0.7494	0.7333	0.7366	0.7366	0.7333
Random Forest	0.7500	0.7514	0.7522	0.7499	0.7833	0.7848	0.7857	0.7833
Gradient Boosting	0.7333	0.7321	0.7321	0.7321	0.7667	0.7701	0.7701	0.7667

4.3.2. Explainable AI in the Context of Research Output Model

This section analyzes the prediction behavior of the best-performing model using three explainability methods: SHAP, Permutation Importance, and LIME. SHAP and Permutation Importance provide global insights into which features most influence the model's decisions, while LIME offers local explanations for individual predictions. Together, they enhance the transparency of the model and support a deeper understanding of its decision-making process.

4.3.2.1. Global Explanations with SHAP and Permutation Importance

To gain a clearer understanding of which variables influence the model's predictions the most, Figure 4.1 presents the top 10 features ranked by mean absolute SHAP values for the best-performing model. The results show that ST slope_1 is by far the most influential predictor, followed by chest pain type_4, sex, oldpeak, and age. These variables have substantially larger SHAP scores than the remaining features, indicating that they contribute the most to the model's predicted risk of heart disease. Clinically, this is consistent with medical knowledge: ST segment changes and typical angina (chest pain type_4) are strong indicators of myocardial ischemia, while age, sex, and ST depression (oldpeak) are well-established cardiovascular risk factors.

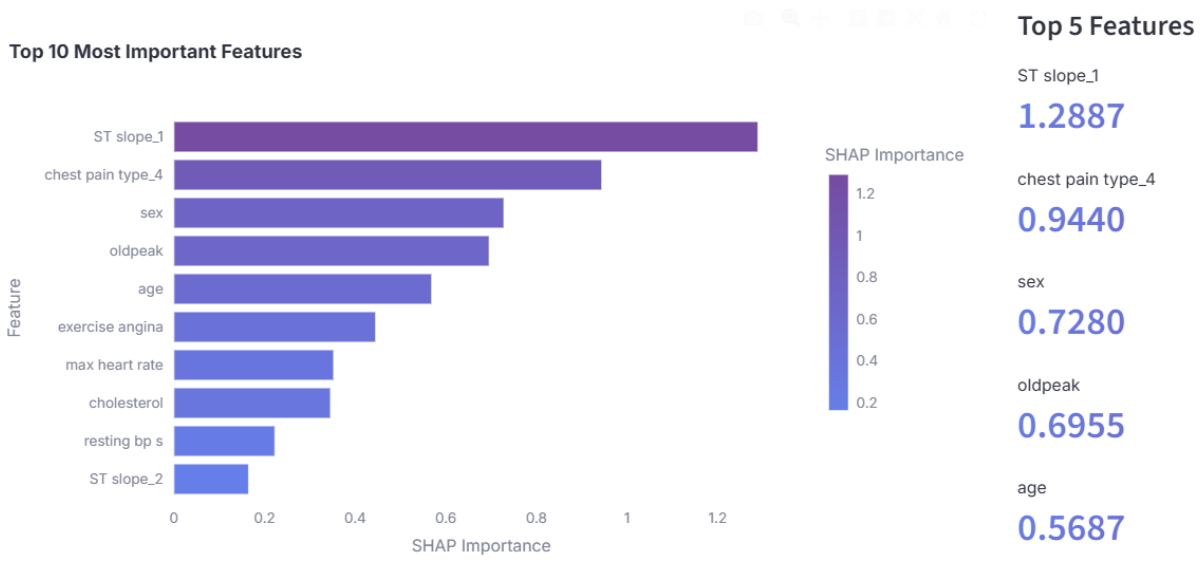

Figure 4.1: Global Feature Importance Based on SHAP Values

Figure 4.2 shows the corresponding permutation importance results, which follow a pattern highly consistent with SHAP. ST slope_1 remains the most influential feature, followed by oldpeak, chest pain type_4, sex, and max heart rate. The strong agreement between SHAP and permutation importance confirms that these variables carry the main predictive signal in the model. Small differences in ranking - for example, oldpeak and chest pain swapping positions are expected due to the different mechanisms behind the two methods.

In addition to ranking the features, the permutation importance plot also provides the standard deviation associated with each importance score across repeated shuffling rounds. Although features such as ST slope_1 and oldpeak show very high importance, their standard deviations indicate that the performance drop caused by shuffling them is not entirely stable. This means the model relies heavily on these features, but the strength of this dependence changes across different perturbations of the data. Such variability is often a sign that these important features interact with other variables, something permutation importance alone cannot fully capture.

For this reason, applying XAI-guided feature engineering remains valuable. Rather than simply identifying which features are important, this approach helps the model make better use of the hidden relationships among them. Creating interaction features from the most influential variables enables the model to learn more complex patterns and reduces its unstable reliance on any single feature. In simple terms, XAI-guided improvement does not replace important features, it helps the model understand how they work together, resulting in more stable and accurate predictions.

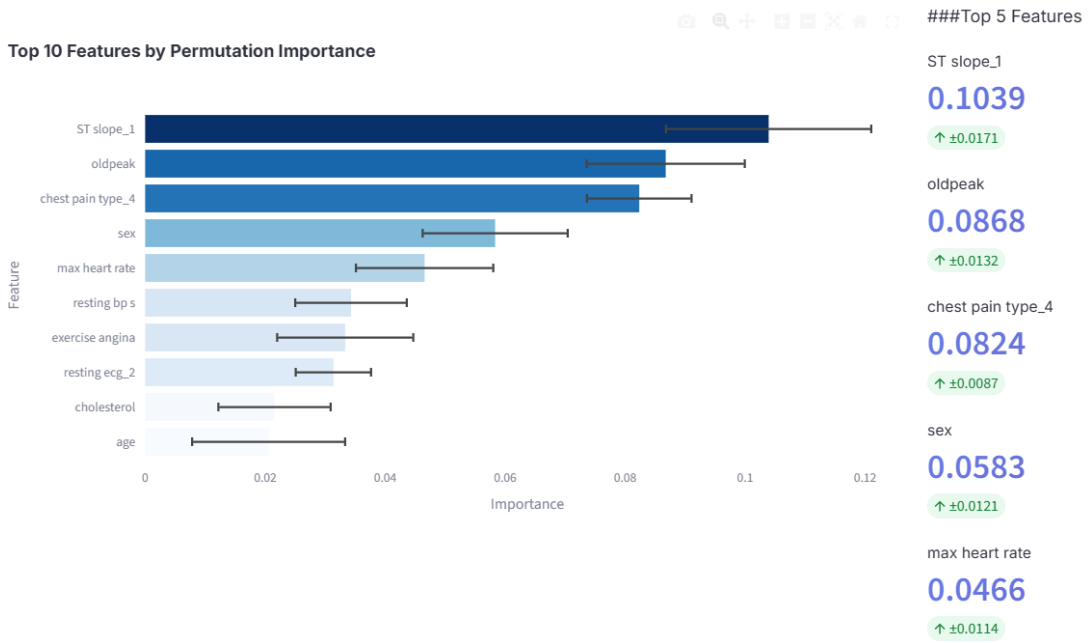

Figure 4.2: Global Feature Importance Based on Permutation Importance

Based on the top-ranked features from Permutation Importance, a focused subset of influential variables—such as ST slope_1, chest pain type_4, oldpeak, sex, and max heart rate—

was selected as candidates for constructing interaction features in the XAI-guided model improvement stage. Permutation Importance directly measures the necessity of each feature for the model by shuffling the feature and assessing the change in accuracy, which helps determine which features are truly important for the model. Meanwhile, SHAP and LIME are used solely for visualization, showing how features contribute to the prediction but not for model improvement. Using Permutation Importance in the model enhancement stage ensures that the selected features are not only statistically informative but also clinically meaningful in the context of heart disease prediction, particularly when there are correlated features.

4.3.2.2. Local Explanations with LIME

In this study, LIME is used to provide local explanations that show how the model makes decisions for individual patients.

First, the user enters complete patient information through the system's form. After completion, the user selects the "Predict & Explain" function, and the system will make a prediction and display the corresponding results, as illustrated in Figure 4.3. Next, the LIME method is applied to explain the predicted results for the specific case, helping to clarify the factors influencing the model's decision.

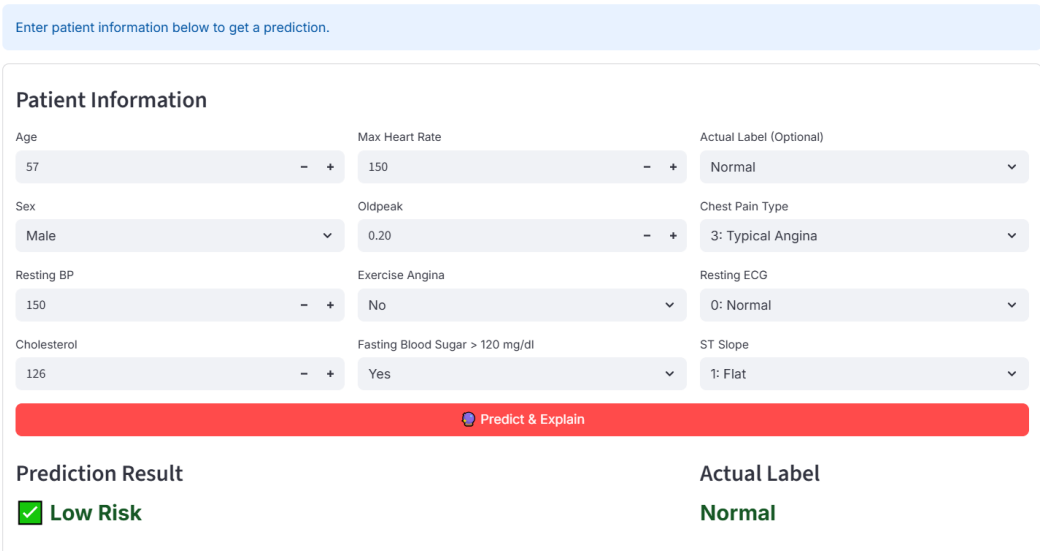

Figure 4.3: Heart Disease Prediction Interface

The Figure 4.4 illustrates the most important characteristics influencing the model's predictive results for this case. Red bars represent factors that push the prediction toward heart disease, while green bars represent factors that reduce the prediction and tend toward no heart disease. The length of each bar reflects the relative contribution of the corresponding characteristic to the model's final decision. Through this, LIME helps users better understand why the model produces the predictive results for each specific case.

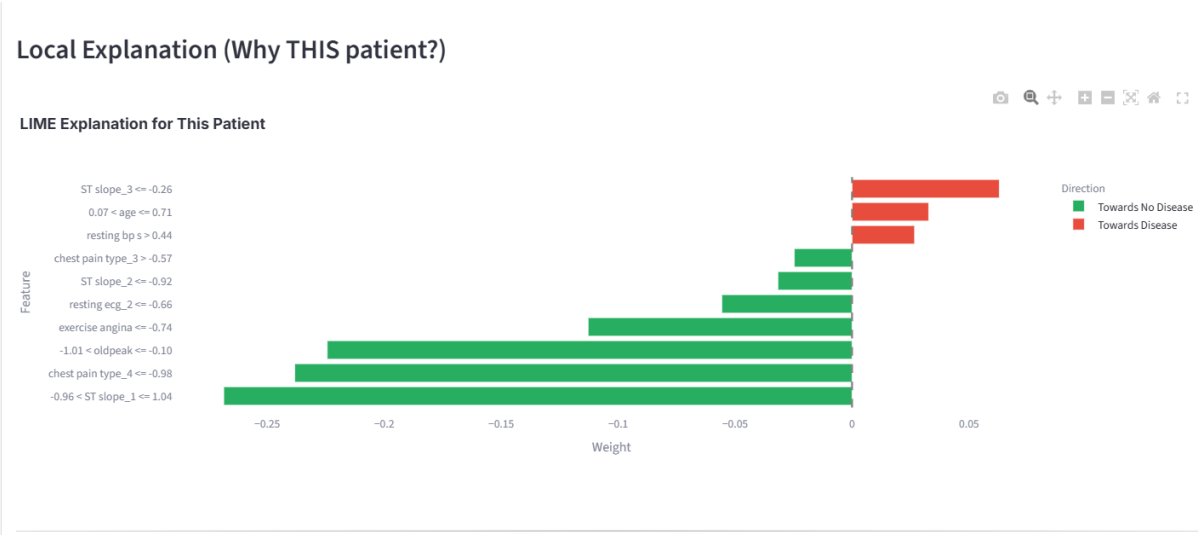

Figure 4.4: Local Explanation Using LIME for an Individual Patient

Based on Figure 4.4, we can explore some meaningful results. In this case, features such as ST slope_1, chest pain type_4, and oldpeak have relatively large negative coefficients, indicating that they play an important role in predicting no heart disease. Some other features, including exercise angina and resting ECG, also reduce the predicted risk, but with a smaller effect. In contrast, features such as ST slope_3, age, and resting blood pressure have positive coefficients, suggesting that they increase the risk of heart disease, although their impact is weaker than that of the risk-reducing features.

Overall, the size of each coefficient shows how important the corresponding feature is in the model’s local decision process. This explanation demonstrates that LIME helps explain individual predictions by highlighting the most influential features and the direction of their effects on the model output.

4.3.3. XAI-Guided Model Improvement

After completing the Feature Engineering (FE) process and performing hyperparameter tuning to obtain the best-performing models, an additional experiment was conducted to evaluate whether XAI-guided feature engineering could further improve performance. The overall experimental workflow is shown in Figure 4.5.

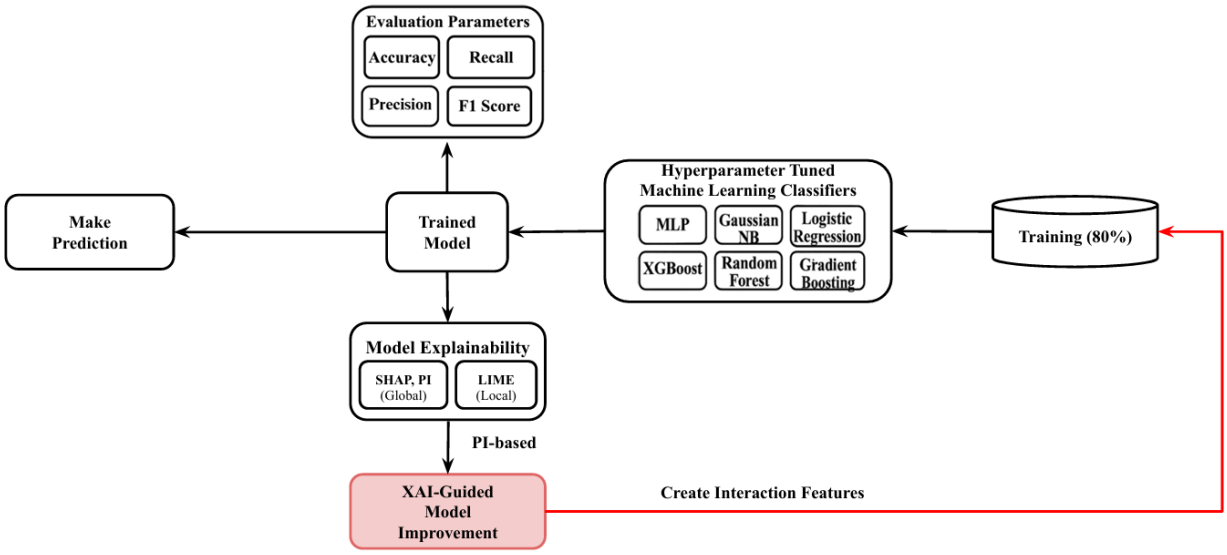

Figure 4.5: XAI-Guided Model Improvement Workflow

In this stage, Permutation Importance was used to identify which features were the most important for the model. This method works by randomly shuffling the values of a feature and observing how much the model's accuracy drops. If the performance decreases a lot, it means that the feature is important. Using this approach, a ranked list of important features was created. From the top important features, new interaction features were generated. These interaction features were created by multiplying the values of two important features, helping the model capture additional relationships that the original Feature Engineering (FE) dataset might not show clearly. These interaction features were then added to the original FE dataset. By combining the original FE features with the new XAI-guided interaction features, an extended dataset was formed, giving the models more information to learn from.

The models were not tuned again. Instead, each model was retrained using the same hyperparameters that had already been optimized earlier. This approach helps ensure that any change in performance comes from the new XAI-guided features, not from changes in hyperparameter settings. After retraining, the new results were compared with the original FE models to see whether the XAI-guided features provided any improvement.

Table 4.9 summarizes the comparison between the baseline models (tuned and trained on the feature-engineered dataset) and the models retrained with XAI-guided interaction features.

Table 4.9: Performance Comparison Before and After XAI-guided feature engineering

Model	**Baseline Accuracy**	**Improved Accuracy**	**Change**	**Change (%)**
MLP	0.9265	0.9216	−0.0049	−0.53%
Gaussian Naïve Bayes	0.8284	0.8431	+0.0147	+1.78%
Logistic Regression	0.8333	0.8235	−0.0098	−1.18%
XGBoost	0.9363	0.9167	−0.0196	−2.09%
Random Forest	0.9216	0.9314	+0.0098	+1.06%
Gradient Boosting	0.9461	0.9363	−0.0098	−1.04%

Overall, the XAI-guided feature engineering step produces mixed effects. Two models, Gaussian Naïve Bayes and Random Forest, benefit from the additional interaction features, achieving relative accuracy gains of +1.78% and +1.06%, respectively. In contrast, the remaining four models (MLP, Logistic Regression, XGBoost, and Gradient Boosting) experience small performance drops. On average, the mean accuracy across all models decreases slightly from 0.8987 (baseline FE) to 0.8954 (XAI-guided), indicating that the feature space obtained from statistical feature engineering is already close to optimal for this dataset.

Importantly, the best overall performance is still achieved by the baseline Gradient Boosting model trained on the FE dataset (accuracy = 0.9461), rather than its XAI-guided variant. This suggests that, while XAI-guided interaction features can improve certain models (especially simpler or more variance-sensitive ones like Gaussian Naïve Bayes and Random Forest), they do not universally guarantee performance gains. Once the core feature set has been carefully refined by statistical methods, adding extra features may not always lead to improvements and may, in some cases, even reduce the model's ability to generalize to new data.

Specifically, for Gradient Boosting, a model that has already been optimized, if it performs well on the dataset after basic Feature Engineering, adding new interaction features from XAI could lead to overfitting or reduce the model's stability, especially if the number of added features is too large or inappropriate. Moreover, the nature of XAI-guided feature engineering needs to be considered: XAI often focuses on creating interaction features based on model insights, but these new features do not always bring improvements to the model. This is especially true for complex and powerful models like Gradient Boosting, where additional features may reduce the model's ability to generalize.

4.3.4. LLM-based Interpretation

Although LIME provides local explanations for individual predictions, its outputs can be hard for non-technical users to understand. Therefore, the LIME explanation for each patient is converted into a clear and easy-to-read description using an LLM to support clinical use. This workflow is showned in Figure 4.6.

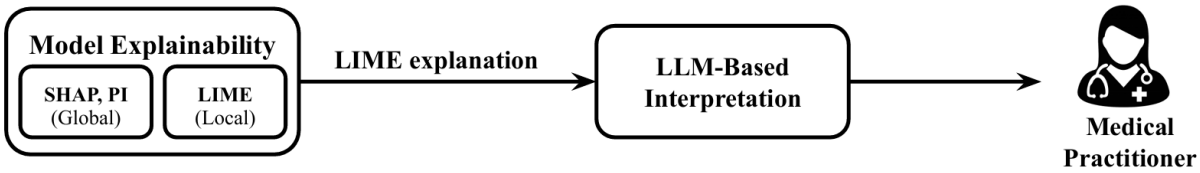

Figure 4.6: Workflow for Interpreting LIME Explanations Using an LLM

In this study, the LLM is used to convert the local explanations generated by LIME into clear and easy to understand for each individual patient. For each individual case, the LLM receives the following inputs:

i. Prediction result: the predicted class (heart disease or no heart disease) and the confidence score produced by the model.
ii. Important features: the key features identified by LIME that influence the prediction for the specific patient.
iii. Contribution scores: the LIME weights showing whether each feature increases or decreases the predicted risk.
iv. Patient input values: the actual values of the patient used for prediction.

Based on these inputs, a predefined prompt guides the LLM to generate simple and medically meaningful explanations. The generated output includes:

i. Explanation of the prediction: why the model produced this result for the patient, based on the LIME features.
ii. Direction of impact: whether each important feature increases or reduces the predicted risk.
iii. Personal recommendations: general and non-diagnostic suggestions based on the patient's risk profile.

This approach helps turn LIME outputs into explanations and recommendations that are easier to understand and more useful for clinical support. An example is clearly shown in Figure 4.7, which presents the LLM-generated explanations for the above patient.

1. Why the model gave this prediction – step-by-step (LIME explanation)

The LIME (Local Interpretable Model-agnostic Explanations) output shows which *features* pushed the prediction **away** from "heart disease" and which nudged it **toward** it.
The numbers in parentheses are the *contribution* (negative = pushes toward "no disease", positive = pushes toward "disease").

Feature (range)	Contribution	Why it matters for this patient
ST-slope 1 (up-sloping) −0.27	**Strong negative** (protective)	An up-sloping ST segment during stress testing is the most reassuring pattern; it is associated with a very low likelihood of obstructive coronary disease.
Chest-pain type = 4 (asymptomatic) ≤-0.98 −0.24	**Negative** (protective)	The rule actually captures the *absence* of the "asymptomatic" category; because the patient's chest-pain type is 3 (typical angina), this rule does **not** fire, so it does not add risk.
Oldpeak between −1.01 and −0.10 −0.22	**Negative** (protective)	The patient's old-peak is **+0.2**, which lies *outside* the risky negative range; the model therefore sees the measured value as essentially normal (very little ST-segment depression).
Exercise angina ≤-0.74 −0.11	**Negative** (protective)	The patient reported **no** angina during exercise (value = 0), satisfying the "no exercise-induced angina" rule, which strongly reduces predicted risk.
ST-slope = 3 (flat) ≤-0.26 +0.06	**Slight positive** (tiny risk)	The patient does **not** have a flat ST-slope, so this small risk factor is irrelevant.
Resting ECG = 2 (ST-T abnormality) ≤-0.66 −0.06	**Negative** (protective)	The ECG is *normal* (code 0), so the model does not see the abnormal ECG pattern that would increase risk.
Age 0.07 – 0.71 +0.03	**Slight positive** (tiny risk)	Age is modestly above the lower bound of the "young" bucket used by the model, adding a minute amount of risk.
Resting BP > 0.44 +0.03	**Slight positive** (tiny risk)	The blood-pressure value is high (150 mm Hg) – this adds a small amount of risk, but it is outweighed by the many protective signals.

Figure 4.7: Example of LLM-Generated Interpretation Based on LIME Output

4.4. Evaluation

Based on the experimental results, the Gradient Boosting model trained on the feature-engineered (FE) dataset achieved the highest performance on the test set, with an accuracy of 0.9461 and an F1-score of 0.9461. Therefore, this model was selected as the final model for the heart disease prediction system. Although XAI-guided feature engineering was applied to further explore potential interactions among important features, the results show that the performance of the Gradient Boosting model did not improve and even slightly decreased after adding interaction features (Accuracy decreased from 0.9461 to 0.9363). This suggests that for a well-optimized model with strong capability to learn non-linear relationships, such as Gradient Boosting, introducing additional XAI-based interaction features may lead to redundant information or reduced generalization ability.

To further assess model stability and generalization, the models were additionally evaluated on two independent datasets: Cardiovascular Disease Dataset and Heart Disease Cleveland. The results indicate that Feature Engineering does not consistently benefit all models. On the

Cardiovascular Disease dataset, Feature Engineering improved the performance of 4 out of 6 models, while the remaining models showed no change or a slight decrease in performance. On the Cleveland Heart Disease dataset, Feature Engineering improved the performance of 5 out of 6 models, demonstrating a more pronounced positive impact. Although the effectiveness of Feature Engineering depends on each model and dataset, the results suggest that this method generally improves or maintains model performance, while also contributing to enhanced stability and generalizability.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1. Conclusion

This study proposed a comprehensive and explainable machine learning framework for heart disease prediction, covering the full pipeline from data preprocessing and exploratory analysis to feature engineering, model training, explainability, and natural-language interpretation. The main objective was not only to achieve high predictive performance but also to improve transparency, interpretability, and practical usability in clinical decision support.

At the early stage, careful data preprocessing and exploratory data analysis were conducted to ensure data quality and feature relevance. Besides, Visualization, statistical testing, and categorical feature analysis helped identify meaningful variables and remove weak or unreliable features. In addition, feature scaling was applied so that all numerical features were on a similar scale. These steps provided a strong foundation for building robust predictive models.

Multiple machine learning models were trained and evaluated using a consistent pipeline. The results show that feature engineering plays an important role in improving or stabilizing model performance, although its effectiveness depends on both the model type and the characteristics of the dataset. In particular, tree-based and simpler models were more sensitive to feature transformations, while highly expressive models were already capable of capturing complex patterns without additional interactions.

To enhance transparency, this study applied three XAI techniques, which is SHAP, Permutation Importance, and LIME to analyze both global and local model's behavior. The explanations consistently highlighted clinically meaningful predictors, confirming that the models rely on medically plausible signals rather than arbitrary correlations.

Furthermore, an LLM is used to explain the results of patient cases in LIME, which helps explain complex numerical results of LIME into meaningful clinical information for users. Therefore, the system enables users to more easily understand the prediction outcomes.

Overall, the proposed framework demonstrates that combining statistical feature engineering, explainable AI, and large language models can produce a heart disease prediction system that is not only accurate but also interpretable, stable, and suitable for real-world clinical decision support.

5.2. Future Work

While this study shows promising results, there are several directions for future research to further improve the proposed system.

First, future work can focus on expanding both the scale and diversity of the datasets. Incorporating larger datasets, data from multiple medical centers, and longitudinal data collected over time would enable the model to learn patterns of heart disease progression. This would improve the model's generalization ability and enhance its reliability when applied to different patient populations.

Second, more comprehensive biological and clinical features can be incorporated into the model. These may include medical imaging data such as echocardiography or full ECG waveforms, more detailed laboratory test results, and information related to lifestyle factors and medical history. Including such features would allow the model to capture a more complete representation of a patient's cardiovascular condition and facilitate the use of multimodal deep learning approaches.

Third, future studies should focus on evaluating the validity and reliability of LLM-generated explanations. This may involve comparing LLM explanations with expert clinical opinions, assessing consistency across similar cases, and identifying potential hallucinations or misleading interpretations. Such validation is important to ensure that the explanations provided by the LLM are trustworthy and suitable for clinical use.

Finally, the XAI–LLM integration layer can be extended into an interactive clinical reasoning assistant. Instead of providing static explanations, the system could allow clinicians

to interact with the model using natural language queries and support counterfactual or “what-if” analyses, thereby enhancing clinical decision-making

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